

The atom and its particles

What is an atom?

Atoms are the building blocks that make up all the matter around us: the ground, the air and all gases, water and all liquids, food, chemical elements, and even our own body. Everything around us, including ourselves, is made of atoms.

! Very important

The atom is the smallest unit of matter that retains the properties of an element. All the atoms known to humanity are organized in the [Periodic Table](#).

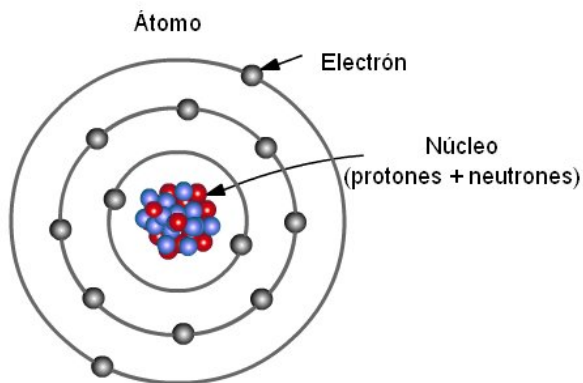
Atoms are extremely small (nanometric scale, between $0,1 \cdot 10^{-9}m$ and $0,5 \cdot 10^{-9}m$): they cannot be seen with the naked eye or with an optical microscope.

i Learn more

Atoms can only be seen with special microscopes such as: [Scanning Tunneling Microscope \(STM\)](#), [Atomic Force Microscope \(AFM\)](#), [High-resolution Transmission Electron Microscope \(TEM/STEM\)](#).

Subatomic particles

Here we show a model of the atom based on the model proposed by [Bohr](#). We say it is a model because the atom is actually much stranger. According to this model, the atom can be drawn like this:



Atoms are made up of three main particles:

Particle	Symbol in notes	Electric charge	Mass (approx.)	Location
Proton	p^+	+1	1	Nucleus
Neutron	n	0	1	Nucleus
Electron	e^-	-1	~ 0	Orbit around the nucleus in the shell (electron cloud).

- Protons and neutrons are in the nucleus.

- Electrons move around the nucleus in orbitals like planets orbiting the Sun.
- Neutrons and protons are much heavier than electrons. Therefore, when calculating mass, we only add protons and neutrons.

Very important numbers

Atomic number (Z)

The **atomic number (Z)** is the number of protons in an atom.

- It defines the element.
- Example: Hydrogen has $Z = 1$. This can be seen in the periodic table.

$$Z = n^{\circ} p^{+}$$

Where $n^{\circ} p^{+}$ = number of protons.

Mass number (A)

The **mass number (A)** is the sum of the number of protons and neutrons. It tells us the mass of a specific atom (remember that electrons are so light that we ignore them for mass).

$$A = Z + n n$$

Where:

- Z = number of protons.
- $n n$ = number of neutrons.

Electric charge (Q)

To calculate the electric charge, you only need to subtract the number of electrons from the number of protons.

$$Q = n p^{+} + n e^{-}$$

If Q is 0, we say the atom is neutral, which means it has no charge.

If Q is positive, the atom has a positive charge.

If Q is negative, the atom has a negative charge.

 Learn more

Video explaining [electric charge](#)

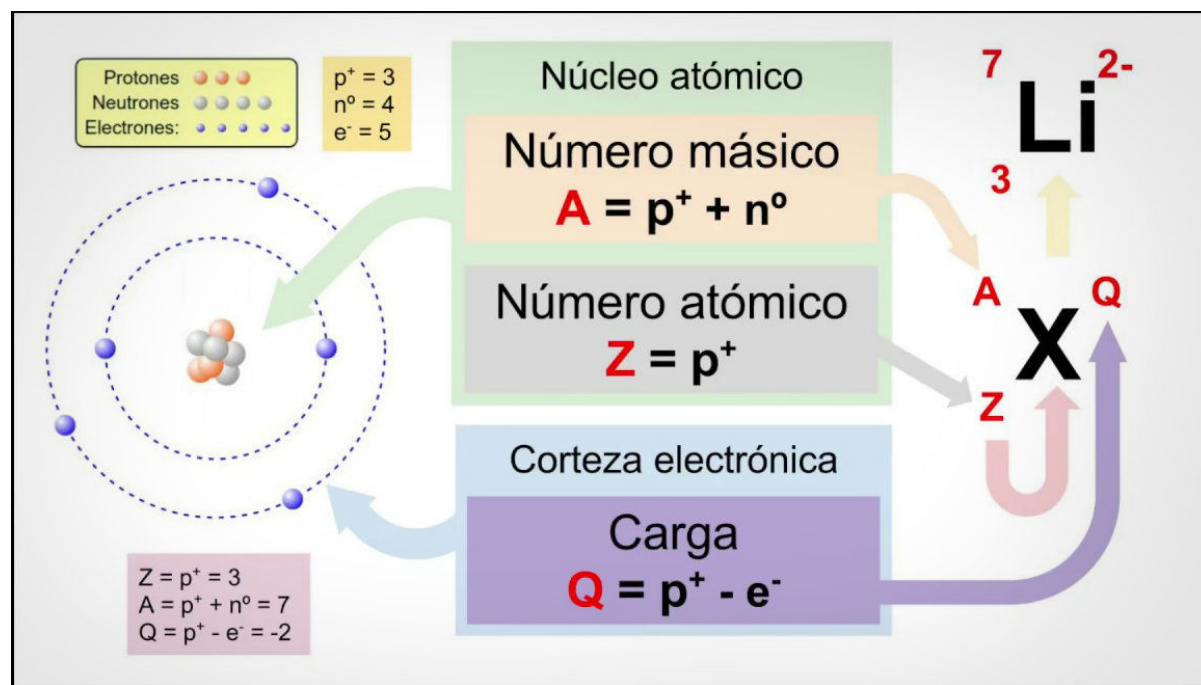
 Very important

Summary of important numbers:

1. Atomic number: $Z = n p^{+}$
2. Mass number: $A = Z + n n$

3. Electric charge: $Q = n p^+ + n e^-$

With these numbers, we usually identify an atom in a very simplified way like this:



Isotopes

Isotopes are atoms of the same element with:

- The same number of protons (Z)
- A different number of neutrons (n)

An example is carbon isotopes: Carbon-12 and Carbon-14. They are written as: *element-A*.

Example of hydrogen isotopes:

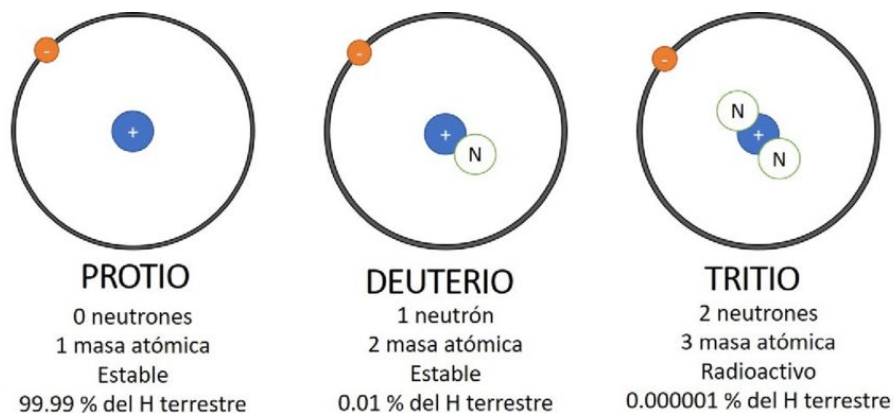


Figure 1: Hydrogen isotopes

Ions

Ions are atoms that have an electric charge, meaning the number of protons is different from the number of electrons. Let's break it down:

- A **neutral atom** is the usual type of atom. This means its electric charge is zero. This happens when we have the same number of protons (p^+) and electrons (e^-). Each positive charge of a proton is balanced by a negative charge of an electron, making the total charge zero.
- An atom that does have an electric charge (that is: $Q \neq 0$) is called an **ion**. Remember this happens when: protons $\neq$ electrons

Types of ions:

- **Cation**: loses electrons $\rightarrow$ positive charge ($Q > 0$).
- **Anion**: gains electrons $\rightarrow$ negative charge ($Q < 0$).

Relationship with the periodic table

- The periodic table is ordered by atomic number (Z).
 - Each element has a unique Z .
 - Elements in the same group have similar properties.
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Summary

! SUPER SUMMARY

- Atoms are made up of protons (+) and neutrons in the nucleus, and electrons (-) in the electron cloud.
 - Z = number of protons.
 - A = protons + neutrons.
 - Q = protons - electrons. Ions have charge:
 - Cation: atom with $Q > 0$
 - Anion: atom with $Q < 0$
 - Isotopes differ in the number of neutrons.
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Exercises

Complete the following table:

Element	Z	A	Protons	Neutrons	Electrons
X	8	16	?	?	?
Y	?	23	11	?	11
Z	17	?	?	18	?

Based on the table above:

1. Calculate the missing values.
2. Indicate whether the atom is neutral or an ion.
3. If it is an ion, indicate its charge.

Challenge question

An atom has: - 12 protons - 12 neutrons - 10 electrons

Answer:

- What is Z?
- What is A?
- Is it an ion?
- What is its charge?